

Roofs and Roof Coverings

Roofs & Their Types:

- Roof may be defined as a covering provided over the top of a building with a view to keep out of rain, snow, sun and wind and to protect the building from the adverse effects of these elements.
- Roof is described as a framework of either timber, steel or concrete on which a covering of thatch, tile, corrugated, asbestos or any other materials of proper insulation is placed.
- A roof basically consists of roof covering materials supported on structural elements installed on the building top.

Functions of Roof:

- To keep out rain, wind, snow and dust.
- To prevent excessive heat loss in winter.
- To keep the interior of the building cool in summer.
- To provide strength, durability and stability for building.
- To protect the building from fire attack.
- To provide resistance to the passage of sound inside the building.
- To provide safety to the occupants.
- To improve the appearance of the building.

Requirement of a roof:

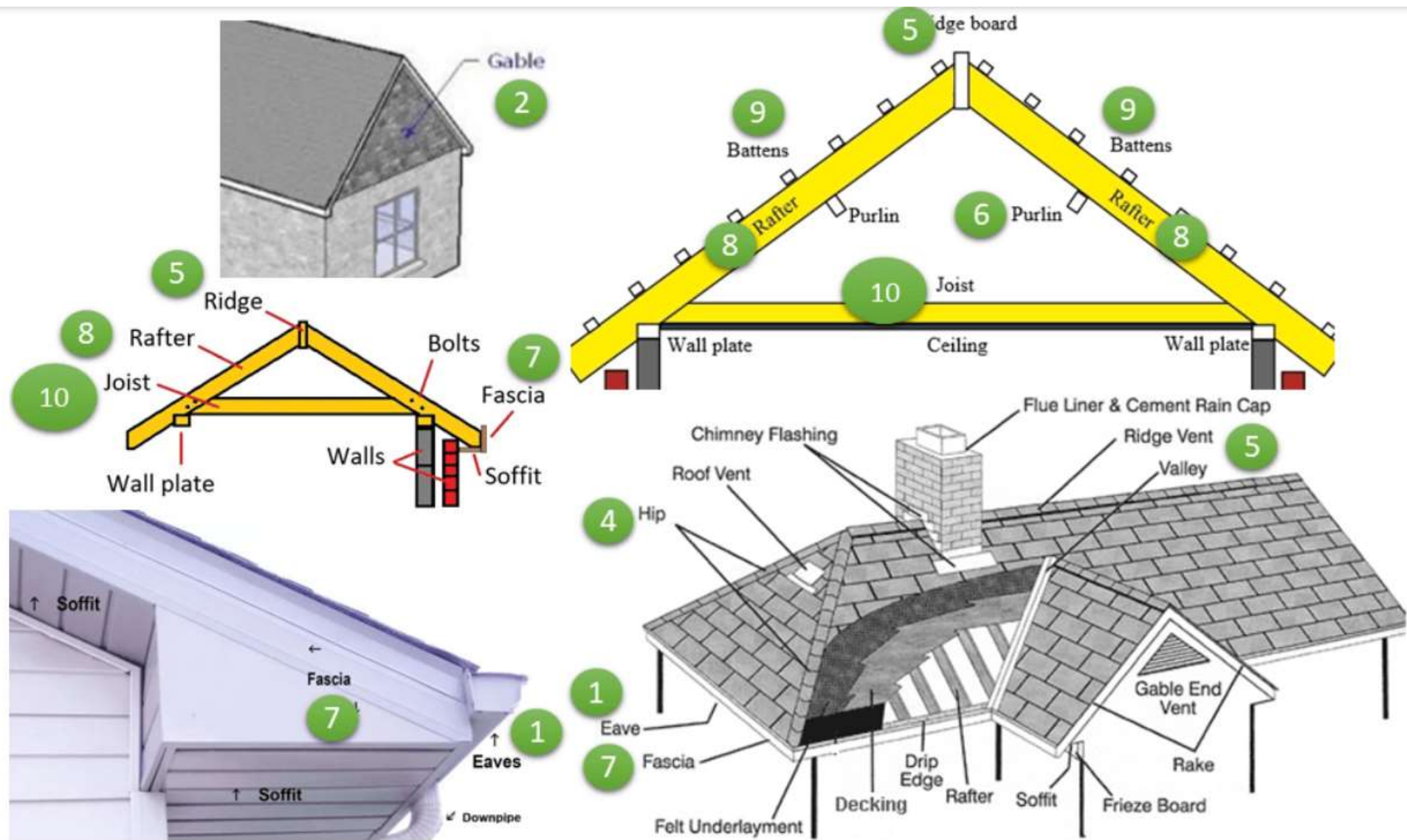
- Should have adequate strength to carry the superimposed loads.
- Should effectively protect the building against rain, sun, wind, etc.
- Should be durable against the adverse effects.
- Should be water proof and should have effective drainage arrangements.
- Should have adequate thermal insulation.
- Should be fire resistant.
- Should provide insulation against external sounds.

Types of Roof:

1. Pitched or sloped roof
2. Flat roof
3. Curved roof

1. Pitched or sloped roof:

- Most common form of roof and is almost always constructed in wood or steel.
- Consists of a system of rafters and purlins arranged in the form of a triangular shaped support known as truss.
- Slope of the roof varies from 10 to 70 degree according to the span, the climatic conditions, the nature of covering materials and other similar factors.
- Generally regarded as the cheapest alternative for covering a structure.

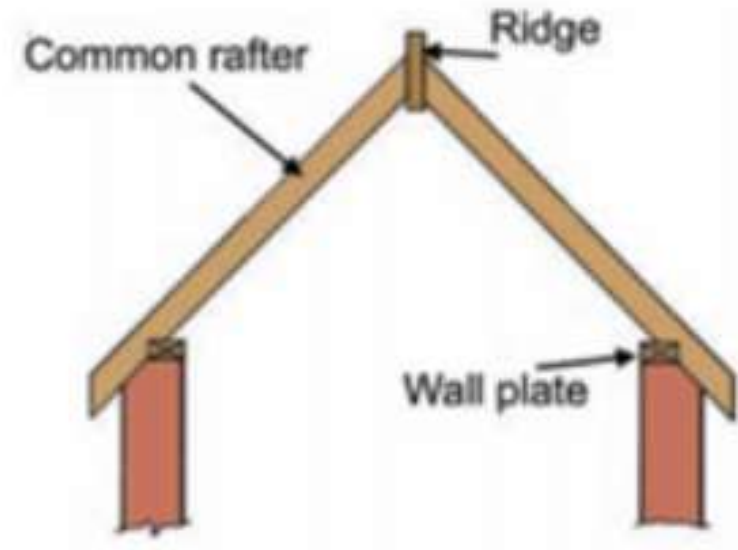


Pitched roofs are of the following types:

- (a) Single roof.
- (b) Double roof.
- (c) Triple or truss roof.
- (d) Mansard roof truss.
- (e) Composite roof truss.
- (f) Combination of king-post and queen-post truss.
- (g) Shelled roof.
- (h) Domed roof.

(a) Single Roof:

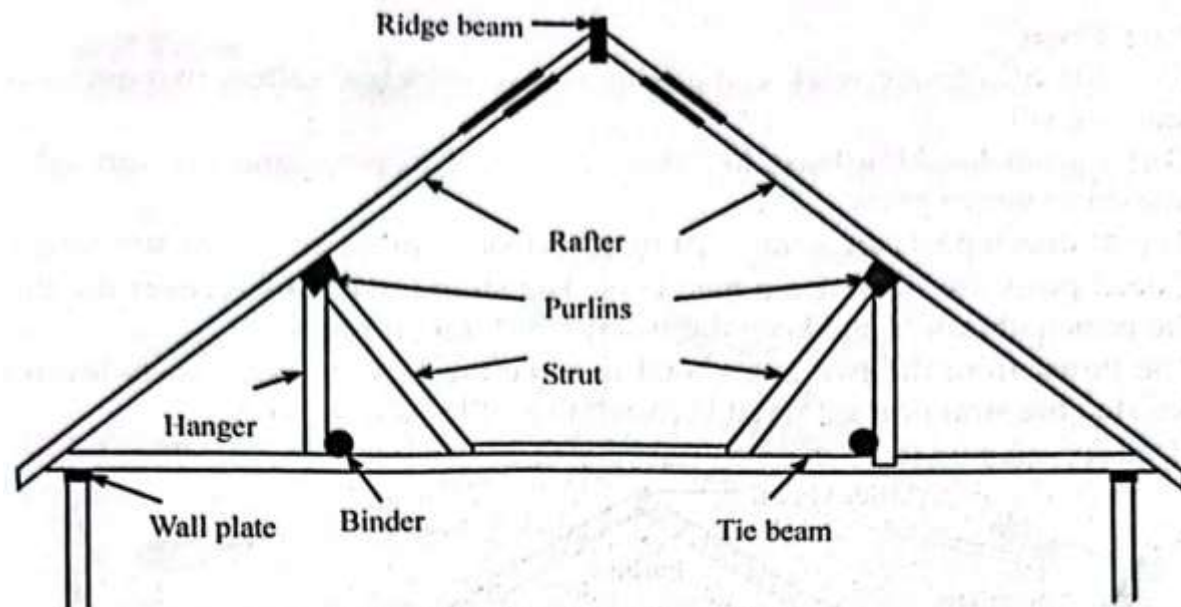
- Consists of only the rafters which are supported at the ridge and the eaves.
- Such roofs are used when the span is limited to 5m otherwise the sizes of the rafters are uneconomical.



Single roof

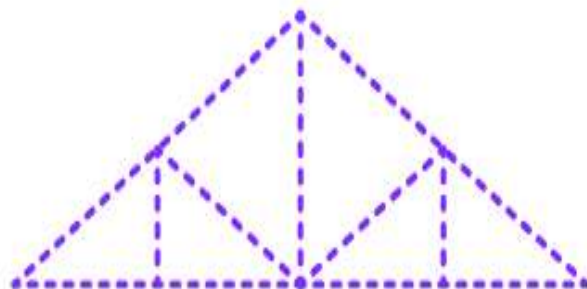
(b) Double or Purlin Roof:

- Consists of two basic elements purlins and rafters.
- Purlins give intermediate support to the rafters and supported on end walls.
- Intermediate supports so provided in the form of purlins reduce the sizes of rafter.

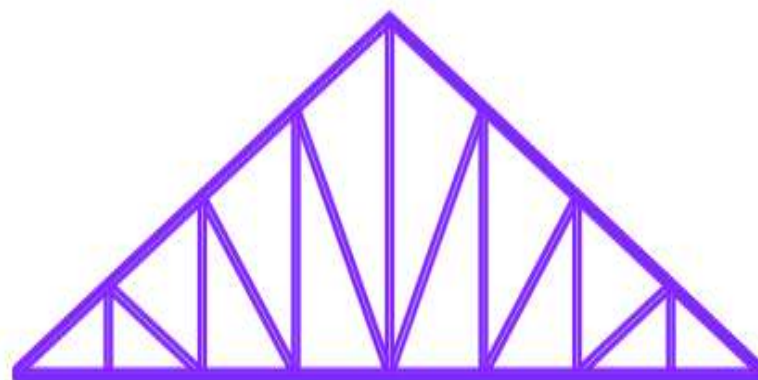


(c) Triple or Truss Roof:

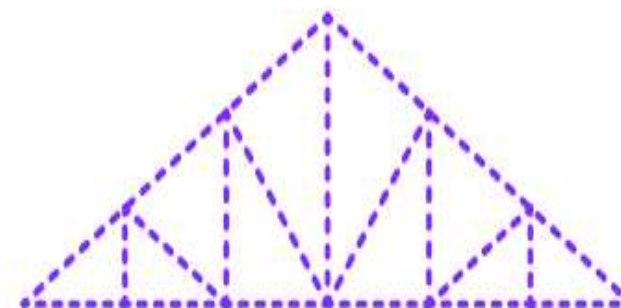
- When the span of the roof exceeds 5m and where there are no inside walls to support the purlins, trusses are provided at suitable interval along the length of the room.
- Spacing of truss is generally limited to 3m for wooden trusses.
- This system of roof consists of three elements.
 - Rafters to support the roofing material.
 - Purlins to provide intermediate support to rafters.
 - Trusses to provide support to the ends of purlins.



Truss - Howe



Truss - Howe, Triple

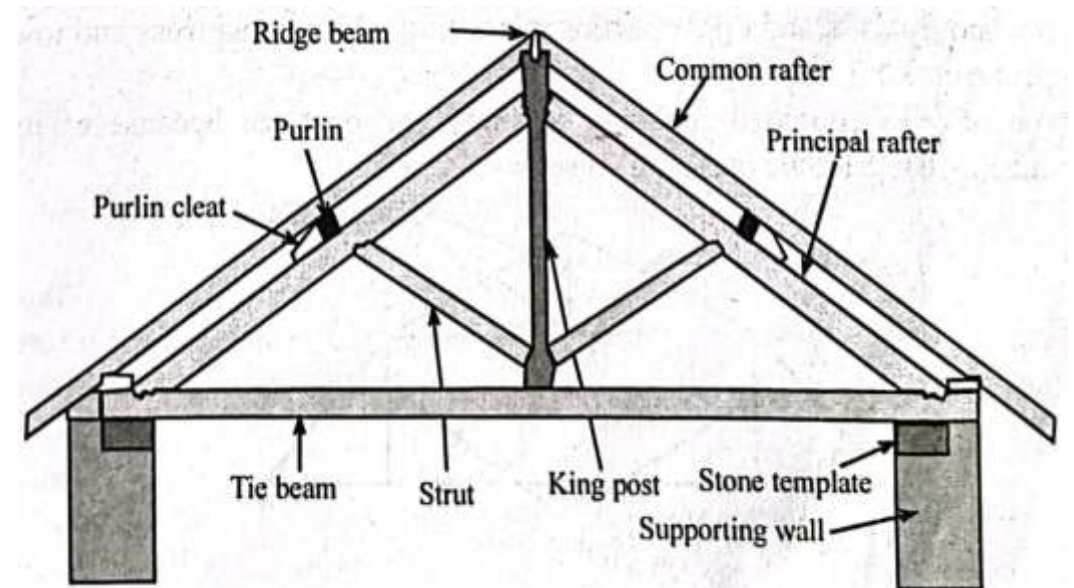


Truss - Howe, Double

Types of Truss:

(i) King Post

- Consists of a frame work comprising of two principle rafters, one tie beam two inclined struts and a king post.
- The trusses are spaced at a distance not more than 3m c/c.
- Commonly used for spans 9 to 12 m.



(ii) Queen Post Truss:

- Consists of a frame work comprising of two principal rafters two queen posts one staining sill.
- Differ from the king truss in having two vertical posts and the vertical posts are known as queen posts.
- Used for span up to 14 m.

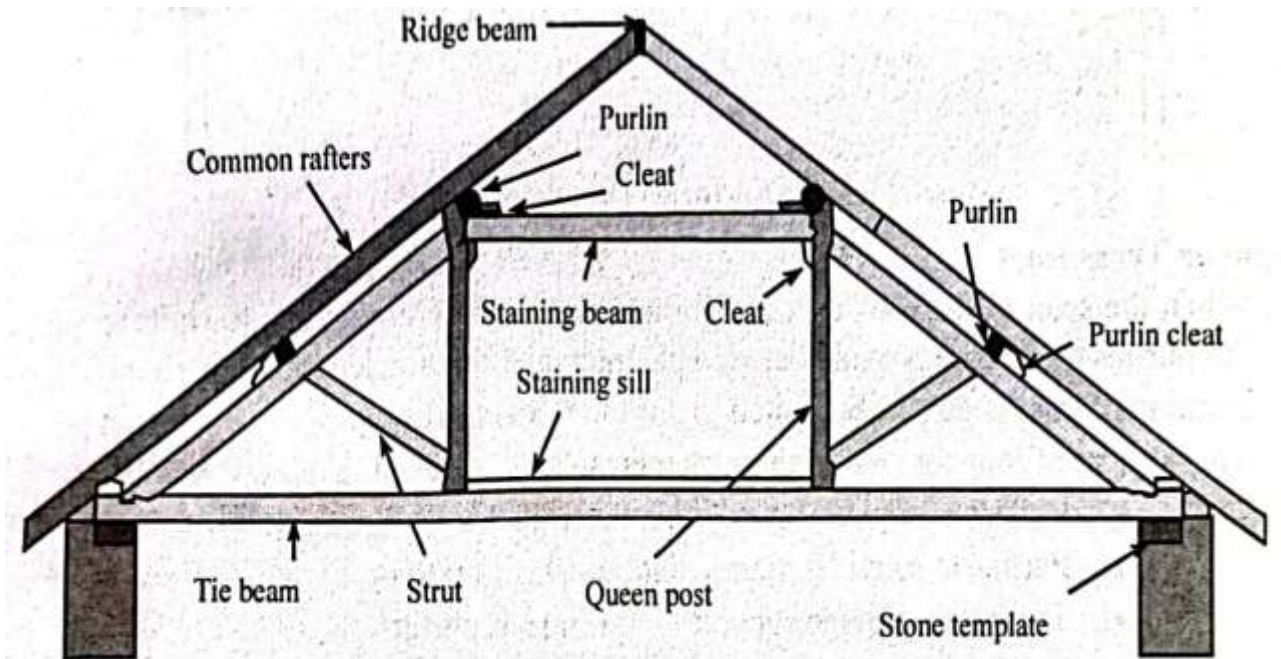
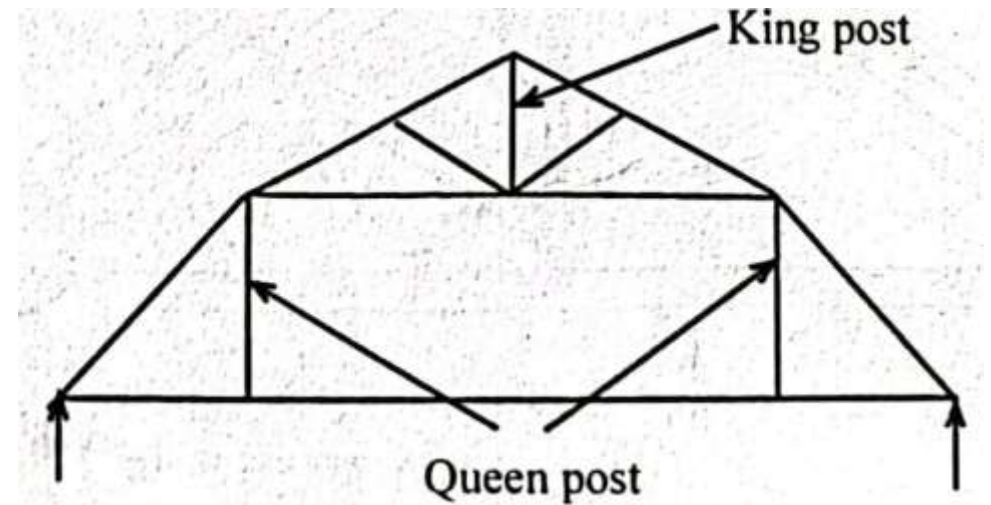


Fig 4.10. Queen Post Roof Truss

(d) Mansard Roof Truss:

- It has two pitches, the lower varying from 60 to 70 degree and upper from 30 to 40 degree.
- This truss is combination of kingpost and queen post truss.
- It is a two storey truss and upper portion consisting of king post truss and lower portion queen post truss.



(e) Composite Roof Truss:

- Roof truss made of two materials such as timber and steel are known as composite roof trusses.
- In composite roof truss the tension members are made of steel while compress members are made of timber.

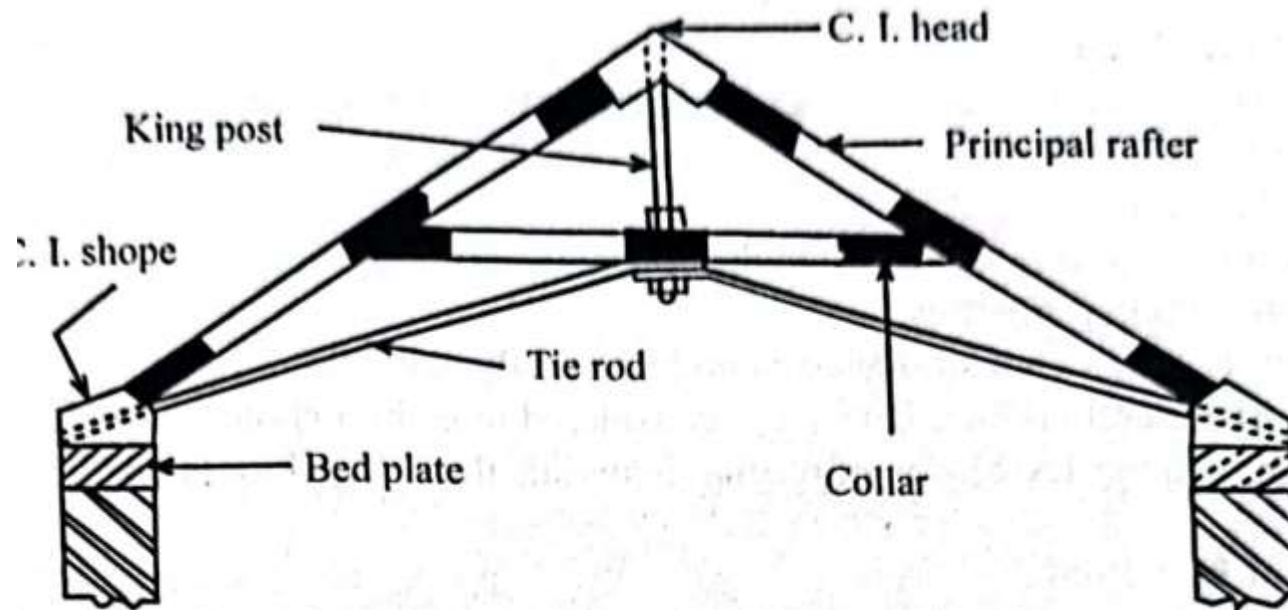
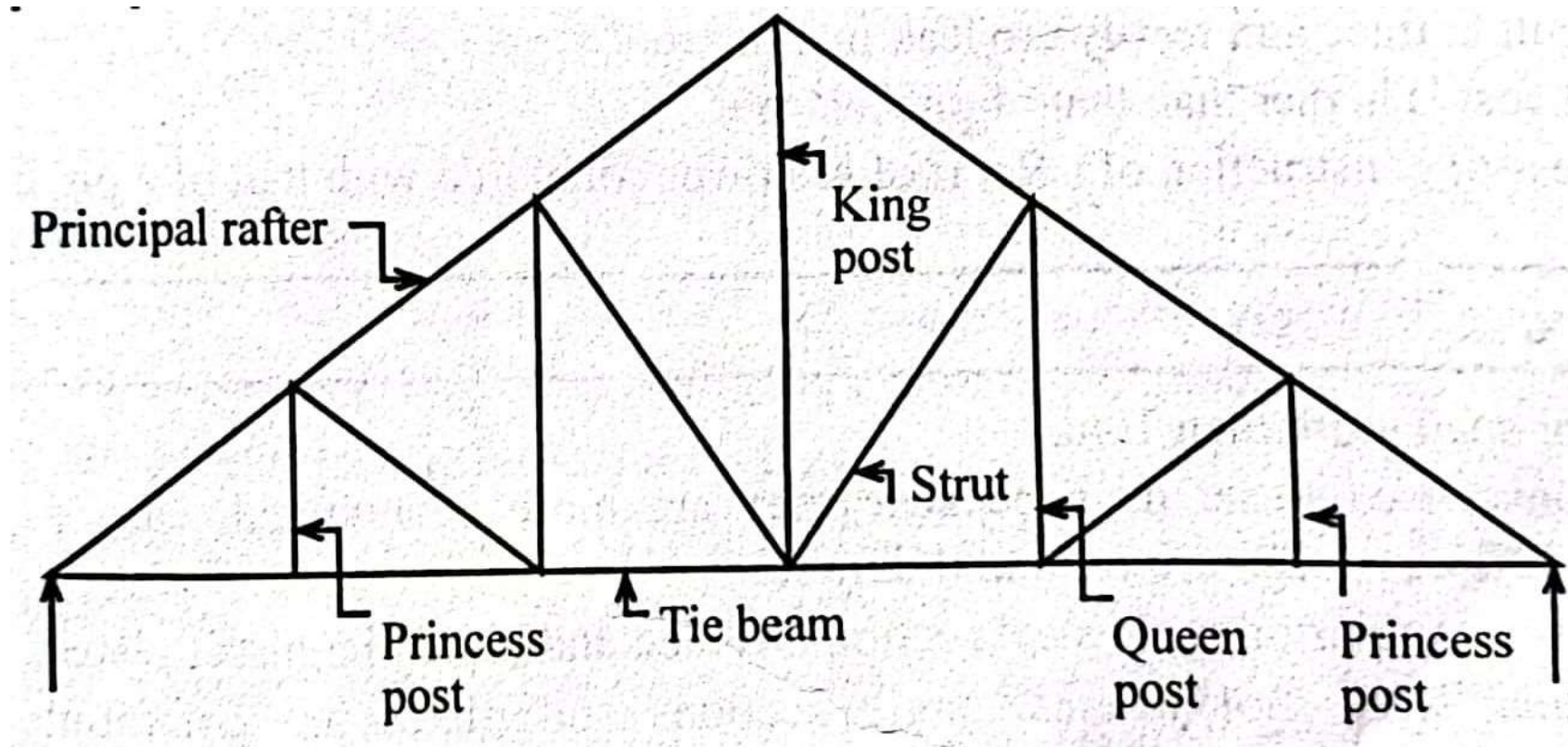


FIG. 4.12. Composite Roof Truss

(f) Combination of King-post and Queen-post Truss:

- It is constructed by combination of king-post and queen-post as shown in figure 4.13 and is suitable for span up to 18m.



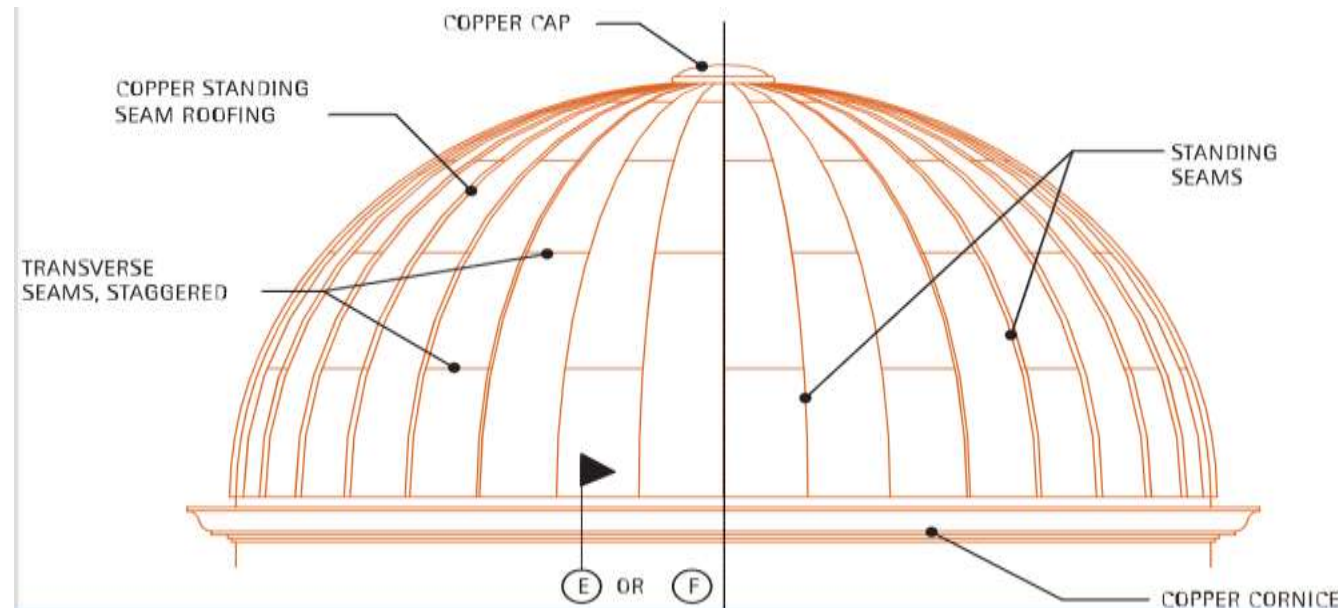
(g) Shelled Roof:

- This is the type of roof that has thin section. It may be constructed in curved surface such as parabola, hyperbola or segment of curve.



(h)Domed Roof:

- This is the curved roof having the roof area more than semicircle. Therefore, dome has largest roof area compare to all types of roofs.
- This can be constructed by the use of R.C.C or arch technology.
- Care must be taken in the construction as it requires high skill.



2. Flat Roof:

- Roof laid at an angle of less than 10 degree to the horizontal is known as flat roof. Flat roof has almost replaced pitched roof after introduction of suitable building materials, although, flat roofs are difficult to construct and expensive in cost.
- Due to various advantages flat roofs are commonly constructed these days.



Advantages of Flat Roof:

- Roof can be conveniently used as terrace, for playing, gardening or for sleeping purposes.
- Better fire resistant.
- Flat roof is considered simplest and architecturally suitable.
- For multistoried building, flat roof is considered to be best choice.

Disadvantages of Flat Roof:

- Span is restricted and can't be used for large spans without introduction of intermediate columns and beams.
- Self weight of flat roof is considerably high which makes the construction expensive.
- Difficult to trace and rectify the leak in a flat roof.
- Initial cost is higher than that of pitched roof.

3. Curved Roof

- A curved roof is a roof structure that has a curved or arched shape instead of a flat or sloping surface.
- It is used to cover large spaces without many internal supports and provides an attractive architectural appearance.
- Example- Dome Roof, Shell Roof



Advantages

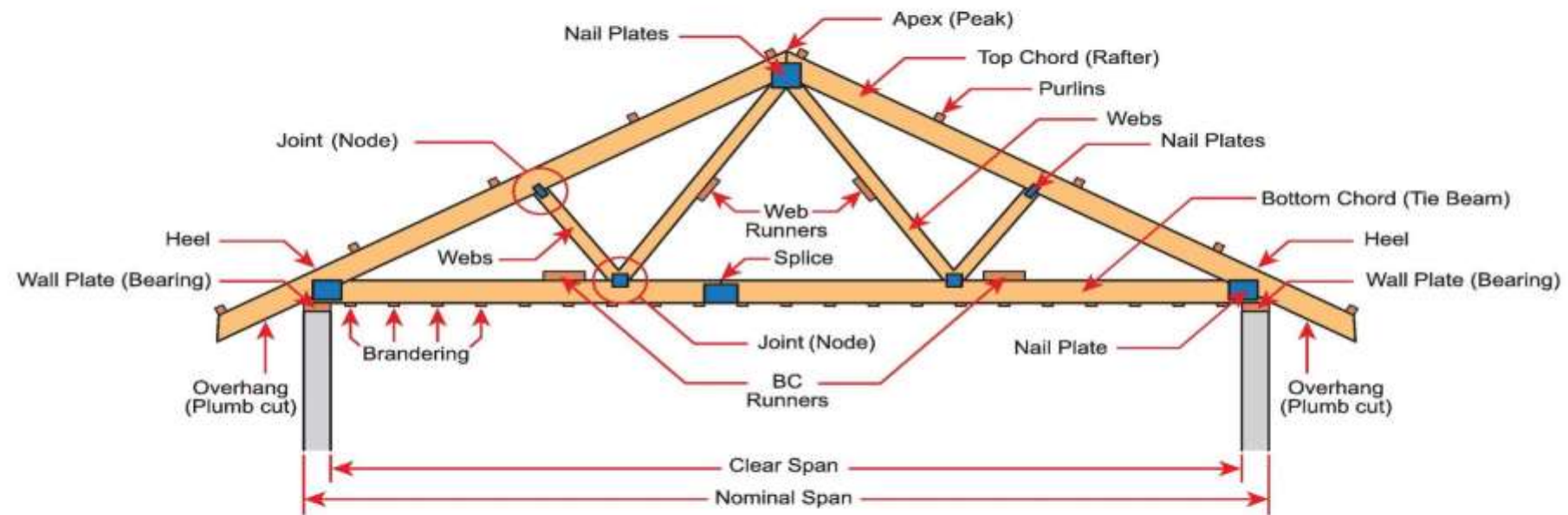
- Attractive and modern appearance.
- Can cover large spans with fewer columns.
- Better wind resistance due to its aerodynamic shape.
- Efficient drainage of rainwater.
- Provides spacious interiors.

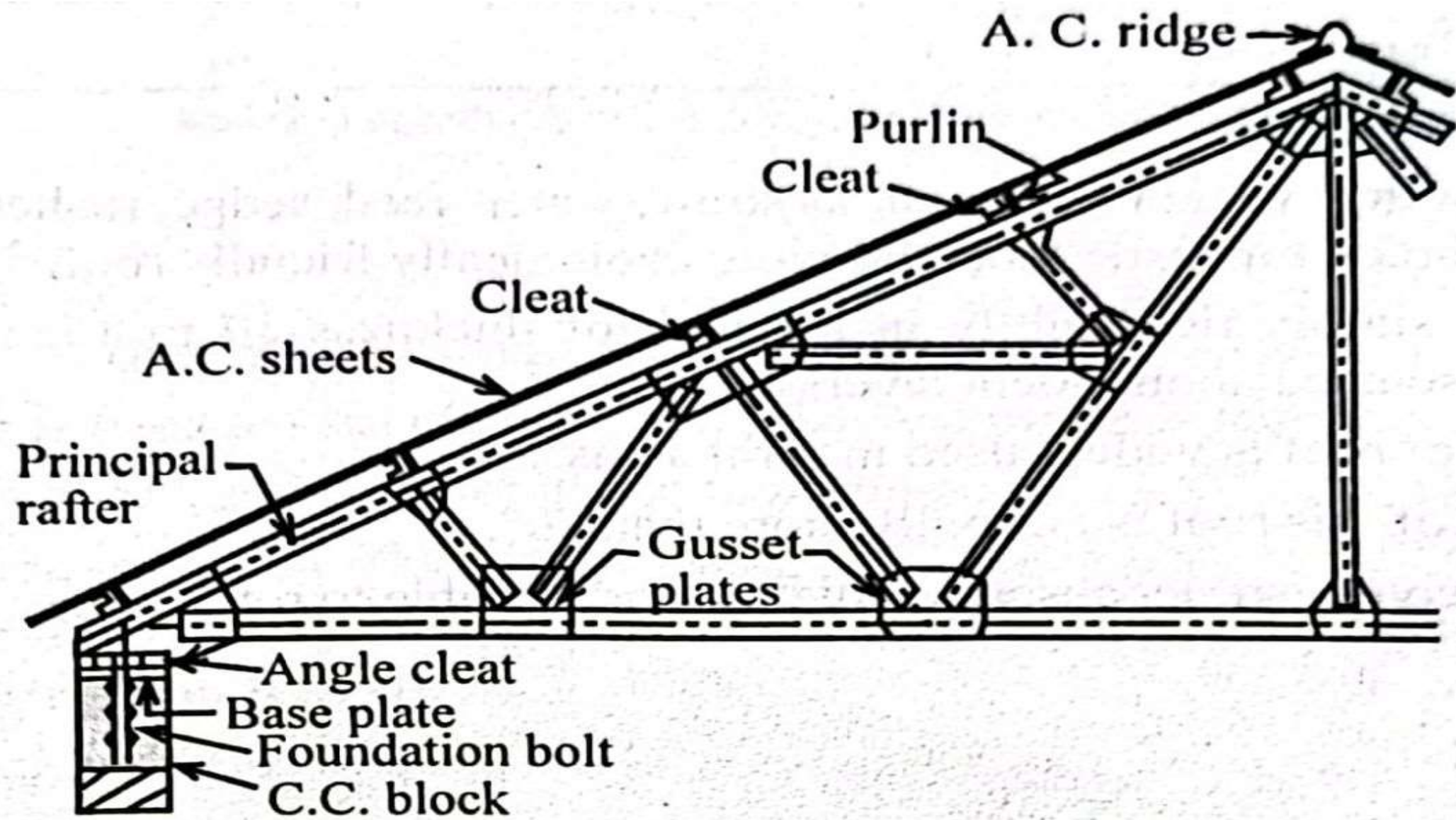
Disadvantages

- More complex design and construction.
- Higher construction cost.
- Requires skilled labor and specialized materials.
- Maintenance and repairs can be difficult.

Steel Truss:

- A steel truss is a framework made of steel members connected together to form triangular units.
- It is used to support roofs, bridges, and other structures by efficiently carrying loads through tension and compression.
- Commonly used these days because they are more economical, easy to construct/ fabricate, fire proof, more rigid and durable.





Main Components of a Steel Truss

1. Top Chord (Principal Rafter)

- Upper inclined member of the truss.
- Carries compressive forces.
- Supports roofing materials.

2. Bottom Chord (Tie Beam)

- Lower horizontal member.
- Resists tensile forces.
- Prevents the truss from spreading outward.

3. Vertical Members

- Connect top and bottom chords.
- Transfer loads within the truss.
- May be in tension or compression.

4. Diagonal Members

- Form triangular patterns.
- Transfer loads between joints.
- Can act in tension or compression.

5. Bracing Members

- Provide lateral stability.
- Resist wind and seismic forces.

6. Panel Points (Joints)

- Locations where members meet.
- Usually connected by bolts, welds, or gusset plates.

7. Gusset Plates

- Steel plates used to connect truss members at joints.
- Distribute forces among members.

8. Purlins

- Horizontal members placed on top of trusses.
- Support roof sheeting and transfer loads to the truss.

Advantages of Steel Trusses

- High strength-to-weight ratio.
- Suitable for long spans.
- Durable and fire-resistant.
- Fast construction.
- Recyclable material.

Disadvantages of Steel Trusses:

- Chance to corrosion.
- Higher initial cost.
- Loses strength in fire.
- Requires regular maintenance.
- Needs skilled labor for fabrication and installation.
- Transportation and erection can be challenging.

Roof Coverings:

1. Corrugated Galvanized Iron Sheets(C.G.I Sheets):

- The rafter is fixed to wall plate and the ridge and purlin is laid on it.
- The C.G.I. sheets are nailed or bolted to purlins.
- Widely used because it is relatively cheap and easy to fix on the roof.



2. Tile Roofing:

- Are made of a ceramic material and is hard and brittle.
- Are manufactured from clay and concrete to a wide range of design and colors suitable for the pitches from 20° to 45°.



3. Slate Roofing:

- Slates is a fine-grained, foliated, homogeneous metamorphic rock derived from an original shale type sedimentary rock composed of clay or volcanic ash.
- The slates are overlapped by 50-75%.
- Long lifespan.
- Heavy and requires a strong roof truss.



Thank You